PiHub

A local Matter-enabled Smart Home Hub

Summer Demo - 2026

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What Changed Since Christmas

Arduino IDE –> ESP-IDF

Espressif toolchain

Required to use ESP-Matter SDK

WSL development

Linux development environment on windows

Needed for ESP-Matter SDK (windows-incompatible)

Python-matter-server

Pi Matter controller

WebSocket used by Home Assistant (industry-standard)

2 Hardware Nodes

Sensor Node - DHT22, TinyML

LED Node - toggle, change brightness, change colour

TinyML Classifier

On-device context classification

CI/CD Pipeline

Two-stack Docker Compose on the Pi

GitHub Actions CI/CD pipeline to GHCR

What is PiHub now?

2 Nodes communicating with python-matter-server via Matter-over-Wi-Fi (IPv6)

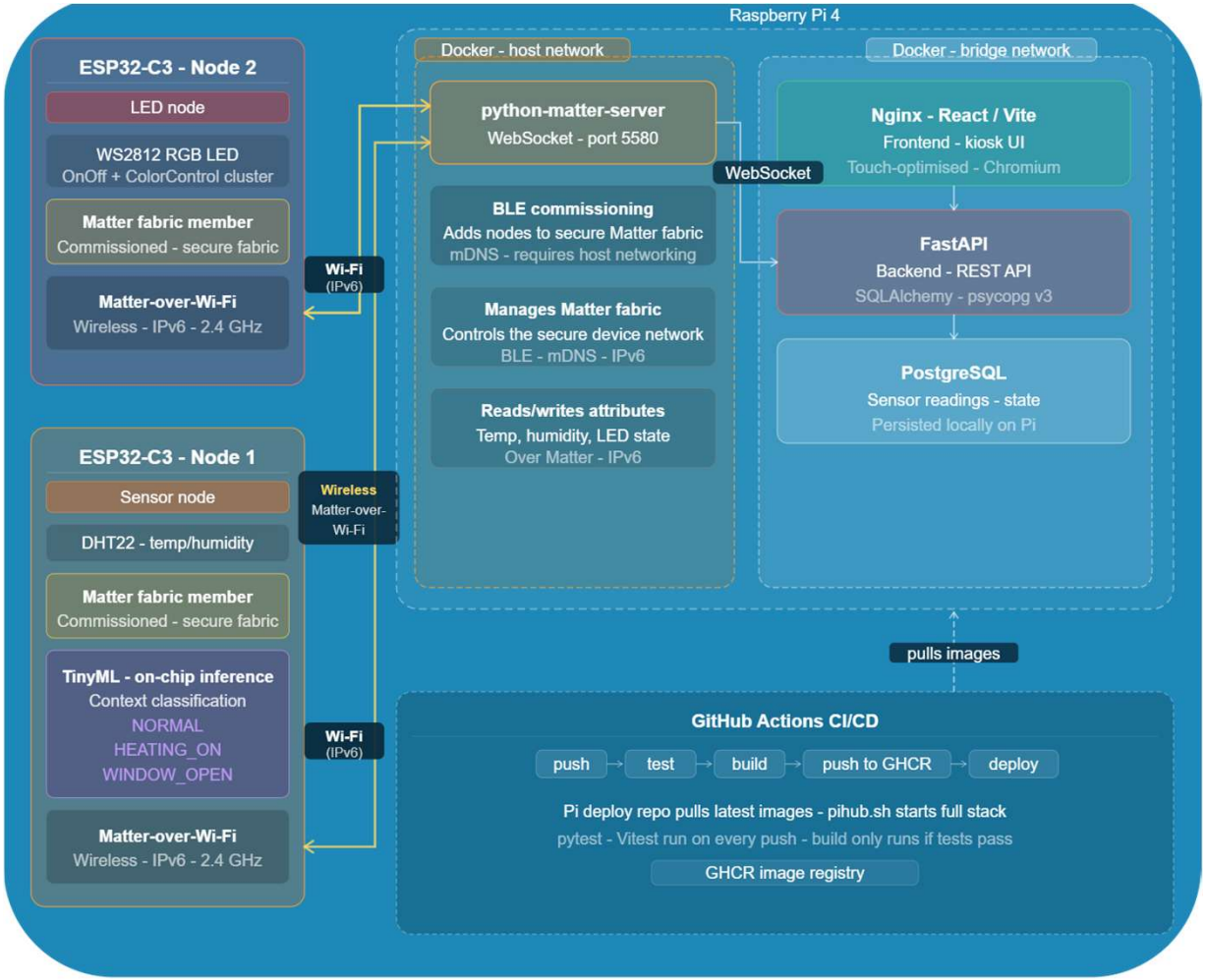
python-matter-server Matter controller

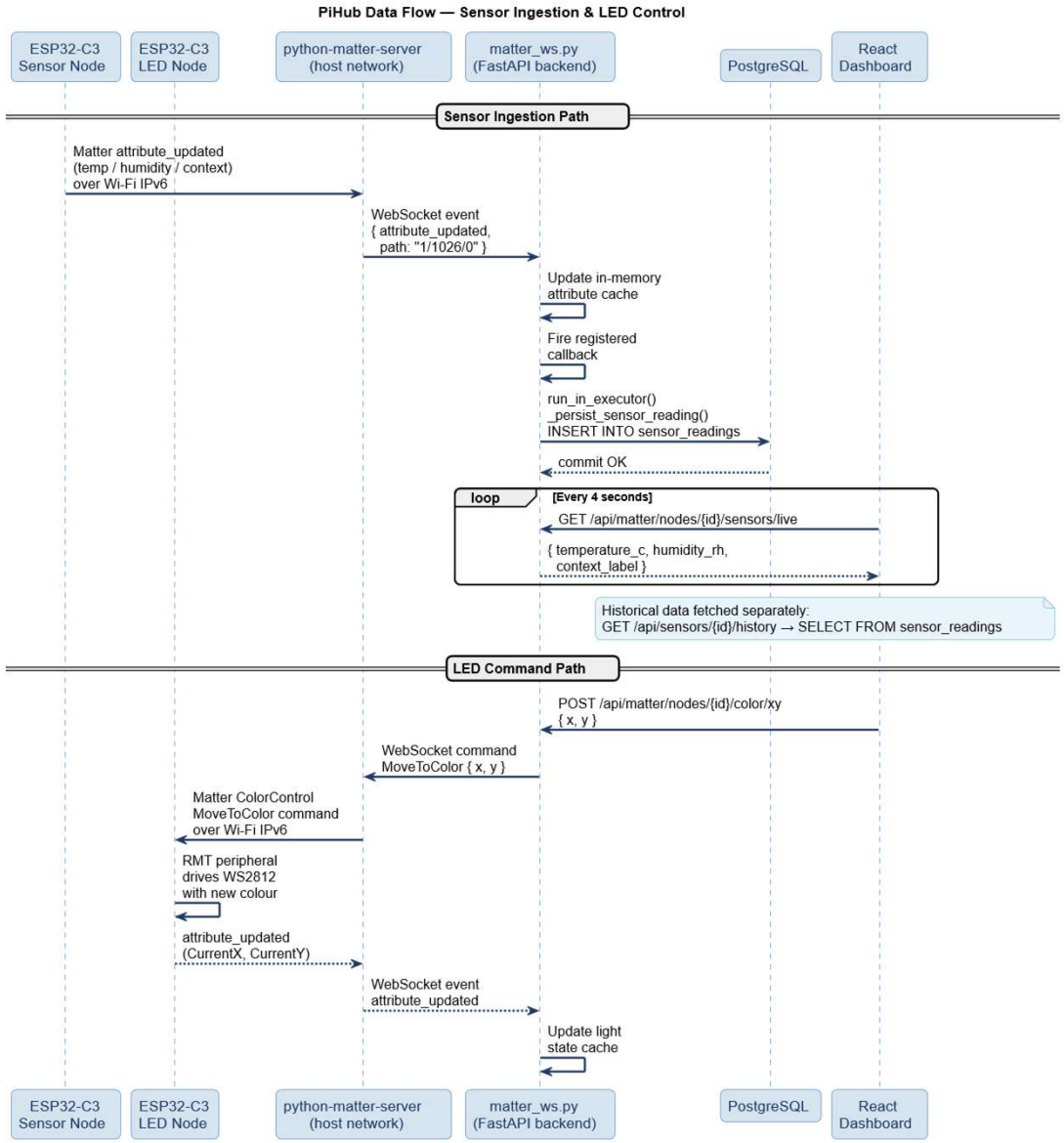
FastAPI talks to Matter controller

Two-stack Docker Compose setup

CI/CD Pipeline

Touchscreen Kiosk connected to Pi




PiHub Data Flow — Sensor Ingestion & LED Control
ESP32-C3 Sensor Node
ESP32-C3 LED Node
python-matter-server (host network)
matter_ws.py (FastAPI backend)
PostgreSQL
React Dashboard
Sensor Ingestion Path
Matter attribute_updated (temp / humidity / context) over Wi-Fi IPv6
WebSocket event { attribute_updated, path: "1/1026/0" }
Update in-memory attribute cache
Fire registered callback
run_in_executor() _persist_sensor_reading() INSERT INTO sensor_readings
commit OK
loop
[Every 4 seconds]
GET /api/matter/nodes/{id}/sensors/live
{ temperature_c, humidity_rh, context_label }
Historical data fetched separately: GET /api/sensors/{id}/history → SELECT FROM sensor_readings
LED Command Path
POST /api/matter/nodes/{id}/color/xy { x, y }
WebSocket command MoveToColor { x, y }
Matter ColorControl MoveToColor command over Wi-Fi IPv6
RMT peripheral drives WS2812 with new colour
attribute_updated (CurrentX, CurrentY)
WebSocket event attribute_updated
Update light state cache
ESP32-C3 Sensor Node
ESP32-C3 LED Node
python-matter-server (host network)
matter_ws.py (FastAPI backend)
PostgreSQL
React Dashboard

TinyML Context Classifier

Train

Keras model (MLP) - feedforward NN

Dummy data

Quantise

Int8 TFLite

C byte array

Deploy

Stored in header file

Compiled into ESP32-C3 DHT22 Firmware

Inference

On-device inference

Temp + humidity combination predicts a class

Outputs

HEATING_ON: 0

NORMAL: 1

WINDOW_OPEN: 2

TinyML Context Classifier

Testing

- Pytest
  - Test files in /test directory
  - Tests split by module
  - AsyncMock - tests WebSocket without a live connection.
- Vitest
  - Tests cover hooks & helper files
  - Coverage low - Screens not tested.

Pytest

```
Name                Stmts   Miss  Cover   Missing
-----------------------------------------------
app/__init__.py         0      0   100%
app/database.py        29      4    86%   54-60
app/main.py           143     12    92%   46-50, 71-73, 138-146
app/matter_ws.py      178     74    58%   107-205, 234, 284-300
app/models.py          17      0   100%
app/schemas.py         33      0   100%
-----------------------------------------------
TOTAL                 400     90    78%
```

Vitest

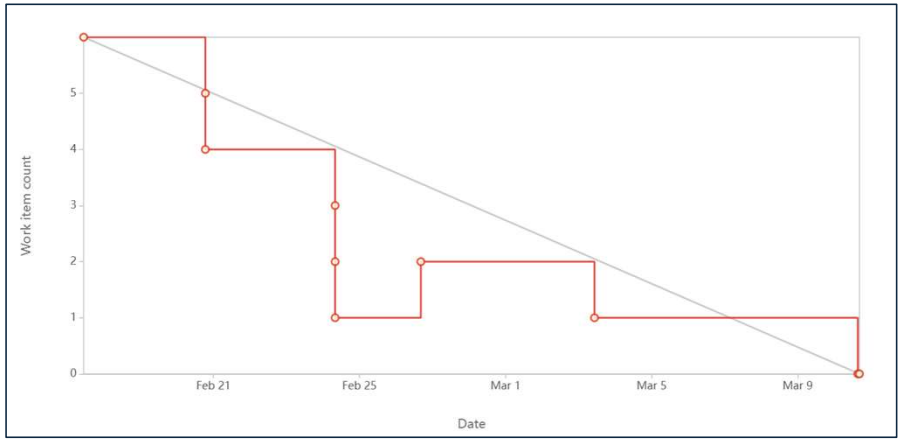
```
File                      | % Stmts | % Branch | % Funcs | % Lines
--------------------------|---------|----------|---------|--------
src/api/                  |     100 |      100 |     100 |     100
ColourHelpers.js          |     100 |      100 |     100 |     100
ChartHelpers.jsx          |    91.3 |      100 |      75 |   88.88
useLedState.js            |   88.13 |    94.44 |   78.57 |   92.15
src/pages/*.jsx           |       0 |        0 |       0 |       0
src/components/*.jsx      |       0 |        0 |       0 |       0
ThemeContext.jsx          |       0 |        0 |       0 |       0
--------------------------|---------|----------|---------|--------
TOTAL                     |   35.91 |    35.76 |      30 |   34.98
```

Project Management - Jira

Ticket work items

Work	Assignee	Reporter
~~PH-142~~ Modify project report template to suit my use case	Unassigned	TM tremain mannion
~~PH-137~~ Create project poster	Unassigned	TM tremain mannion
~~PH-141~~ Take screenshot of webpage dashboard	Unassigned	TM tremain mannion
~~PH-140~~ Take picture of project setup and add to poster	Unassigned	TM tremain mannion
~~PH-139~~ Design poster layout	Unassigned	TM tremain mannion

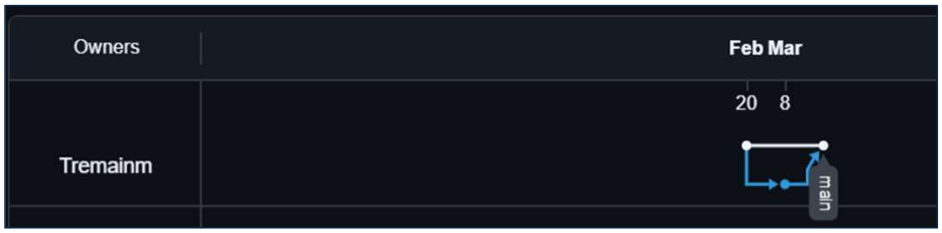
Sprint burndown chat



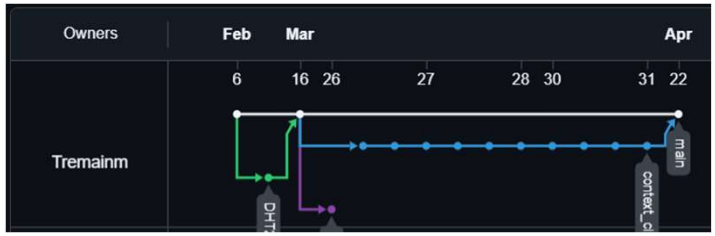
Project Management - GitHub

PiHub App GitHub

LED ESP32-C3 Firmware GitHub



DHT22 ESP32-C3 Firmware GitHub



Project Management - OneNote

Weekly standup

Sem 2, Week 10 - Hardware standup 24/03/2026

Tuesday 24 March 2026 10:06

Tremain:
last week: got all 3d printed parts, modified 18650 battery holders by adding positive and negative terminals for the battery. Installed tp4056 module on battery holder and soldered battery connectors to tp4056. solder tp4056 output wires to power esp32s. Also, containerised project - used deployment repo to pull images from ghcr, and switched from sqlite to postgreSQL.
This week: start working on TinyML - need to gather dht22 readings and create a training script to train a tflite (tensorflow lite) model with.
Blockers: none

Natalie:
Last week: I finished the admin page , and tested the scanning more than one esp 32 in a room- by using dummy data. I couldn't use my other laptop to actually test it but it works.
This week: I am working on the users ui and hope to finish it this week.
Blockers: None

Eoin:
last week: I set up the gesture training side of my project. I added a training mode that records MediaPipe hand landmark points into a CSV file, so I can build my own dataset instead of saving pictures.
this week: I trained the gesture model using that CSV. I trained a Random Forest and saved the trained model as gesture_model.pkl. Next I'm going to load that model into the live backend so the trained gestures work in real time on the overlay.
Blockers: None

Weekly log

Retraining

Tuesday 31 March 2026 21:07

Training Data Issue and Retraining
With everything wired up end to end, the model was classifying incorrectly at realistic room temperatures. A reading of T=26.6°C H=45.2% was being called NORMAL with 0.88 confidence despite clearly being in heating territory. The scatter plot of the original training data explained why.

The original NORMAL class was collected at around 18°C, and HEATING_ON only went up to about 24°C. The room at 26.5°C was outside the entire training distribution so the model had never seen anything like it so it made its best guess, and NORMAL won because it was the nearest cluster boundary.

The fix was to switch to synthetic data that covers a realistic and well-separated operating range based on my typical home temperatures. Rather than trying to recreate specific physical conditions, numpy was used to generate Gaussian-distributed readings centred on physically sensible values for each class:

```
import numpy as np
import pandas as pd
np.random.seed(42)
N = 500
# NORMAL — typical Irish indoor room
normal_temp = np.random.normal(loc=21.0, scale=0.6, size=N)
normal_hum  = np.random.normal(loc=55.0, scale=2.0, size=N)
# HEATING_ON — radiator nearby, temperature elevated, air dries out
heating_temp = np.random.normal(loc=27.0, scale=1.0, size=N)
heating_hum  = np.random.normal(loc=40.0, scale=2.0, size=N)
# WINDOW_OPEN — cold damp outside air coming in
window_temp = np.random.normal(loc=15.0, scale=1.0, size=N)
window_hum  = np.random.normal(loc=70.0, scale=2.5, size=N)
```

The scatter plot confirmed clean separation between all three classes with no overlap, and the closest class boundaries had clear gaps: NORMAL max temperature 23.3°C vs HEATING_ON min 24.1°C, and NORMAL min 19.1°C vs WINDOW_OPEN max 18.1°C:

Prompt log

Sem2, Week 7: 02/03/2026

Tuesday 10 February 2026 12:19

1. I have a React hook called useLedState that polls a Matter cache every 4 seconds. When a user sets a colour on an RGB LED, the colour drifts on every poll because xyToHex conversion is lossy. By the time they adjust brightness, the colour being re-sent is already wrong. How do I prevent the poll from overwriting the user's chosen colour after they explicitly set it?
2. I need to check what colour control commands my ESP32-C3 Matter light device actually supports. It seems like Hue/Saturation might not be available. How do I dump the ColorControl cluster attributes and accepted commands from python-matter-server over WebSocket to see what the device actually supports at the protocol level?
3. My Matter RGB bulb resets its colour to white whenever I send a brightness command. I can see from the attribute dump that LevelControl accepted commands includes command 0 (MoveToLevel) and command 4 (MoveToLevelWithOnOff). What is the difference between these two commands in terms of how they interact with the ColorControl cluster and ColorMode attribute?
4. I set up a Python script to watch attribute_updated events from python-matter-server for my ESP32-C3 light node. When I change brightness, only 1/8/0 (CurrentLevel) fires. No ColorControl events fire at all after brightness, not even a ColorMode change. I have backend code that re-sends MoveToColor after brightness and the logs show it firing correctly, but the device reports nothing. What does it mean when a Matter device accepts a command with HTTP 200 but fires no attribute_updated events in response?
5. I have been trying to fix a colour reset bug on an ESP32-C3 Matter light for a long time. Every fix I try from the backend and frontend sends the correct commands but the device silently ignores them after a

AI Declaration

- AI tools (Claude, ChatGPT) were used to clarify concepts, accelerate development, assist with React code, debugging Matter integration issues, and code understanding. All final implementation and design decisions are my own.